WHAT DOES GPT-2 ACTUALLY READ?

Interpreting GPT-2 with AttnLRP

AIAA 4051 Final Project

GPT-2 Small, SQuAD_v2, SciQ

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*AttnLRP shows **which tokens** GPT-2 relies on and **how fine-tuning changes** that reliance.*

1 INTRODUCTION

GPT-2 can answer correctly while relying on hidden token cues, not explicit reasoning.

AttnLRP traces whether predictions depend on the question, context, or answer boundary.

2 WHY LRP?

What is it?

Attention-aware LRP tracing predictions through attention and feed-forward layers.

Why use it?

Gives signed relevance — separating supporting from suppressive tokens.

3 METHOD

1. Token Attribution
2. Fine-Tuning Shift
3. Layer Relevance

TASK 1: TOKEN RELEVANCE

We apply LRP to GPT-2 on SQuAD_v2 prompts and compute token relevance

(a) Positive evidence highlights answer-relevant entities

Question: Who was the Norse leader? Answer: Rollo

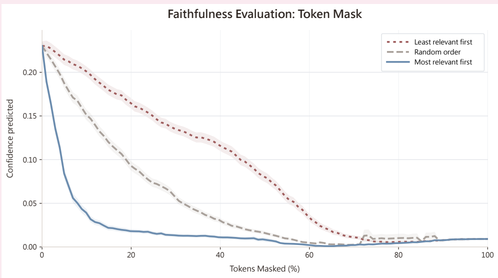
Who | was | the | Norse | leader | ? | Rollo

(b) Negative evidence can suppress confidence

Question: Who wrote the novel 1984? Answer: George Orwell

Who | wrote | 1984 | ? | Answer: | George | Orwell

-1 1 Relevance

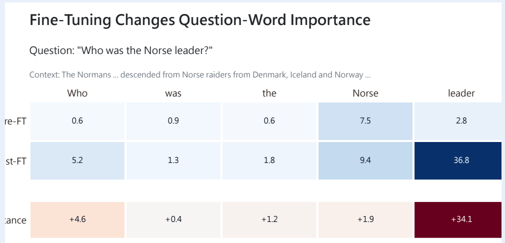


FAITHFULNESS CHECK

Tokens masked by relevance order; curve is non-monotonic since removing suppressive tokens can raise confidence.

TASK 2: FINE-TUNING EFFECT

Using sequential fine-tuning, we compare models on SQuAD_v2 prompts.

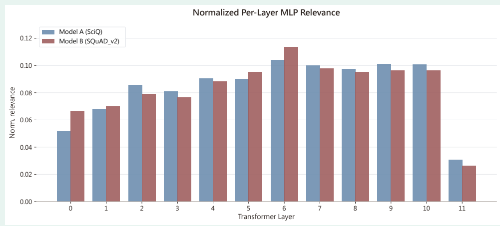


KEY TAKEAWAY

Fine-tuning increases reliance on question-side and answer-boundary cues while reducing background context.

TASK 3: LAYER LRP

We train SciQ and SQuAD_v2 models separately, then compute per-layer MLP relevance over prompts.



KEY TAKEAWAY

Both models show a similar layer profile, with only small task-specific differences near the final layers.

7 KEY FINDINGS

Token relevance captures meaningful answer evidence.

Fine-tuning concentrates attribution around QA cues.

Layer relevance differs by task domain.

8 LIMITATIONS

AttnLRP explains one token at a time — cross-sentence or multi-step reasoning is not captured.

Results are checkpoint-specific; different fine-tuning regimes or out-of-domain inputs may yield different attributions.

9 CONCLUSION

AttnLRP links GPT-2 outputs to token evidence, fine-tuning shifts, and layer usage — showing what the model reads, how training changes it, and where relevance lives.